

Nuclear Talent Course on Quantum Computing for Nuclear Physics, ECT*, June 16- July 4, 2025



I. REPORT

The present document aims at giving a summary, with background information as well, of the Nuclear Talent Course on Quantum Computing for Nuclear Physics. The course was held at the premises of the European Center for Theoretical Studies in Nuclear Physics and Related Areas (ECT*), Trento, Italy during the period June 16 to July 4, 2025.

A. Motivation behind the school

For nuclear theorists, the overarching challenge is to develop a comprehensive description of nuclei and their reactions, grounded in the fundamental interactions between the constituent nucleons with quantifiable uncertainties to maximize predictive power. As experimental frontiers have shifted to the study of rare isotopes, the predictive power of successful phenomenological approaches like the shell model and density functional theory is challenged by the scarcity of nearby experimental data to constrain model parameters. Therefore, it is expected that few-body and many-body methods will play an increasingly prominent role to help improve the predictive power of such “data driven” methods as experiment moves deeper into largely unexplored regions of the nuclear chart.

To understand why nuclear matter is stable, and thereby shed light on the limits of nuclear stability, is one of the overarching aims and intellectual challenges of basic research in nuclear physics. To relate the stability of matter to the underlying fundamental forces and particles of nature as manifested in nuclear matter, is central to present and planned rare isotope facilities. From a theoretical standpoint, this involves understanding how the basic building blocks of Nature interact and conspire to build up atomic nuclei as we know them, with the aim to understand what makes visible matter stable. The theoretical efforts span from methods like lattice quantum chromodynamics, via effective field theories to many-body theories applied to atomic nuclei and infinite nuclear matter. All these methods

rely on theoretical approximations whose applicabilities are often limited by the dimensionality of the specific problem being studied. In recent years, there has been considerable progress in developing quantum-computing algorithms applied to quantum many-body systems, with the hope to circumvent many of the classically intractable problems.

This Nuclear Talent school aimed at bringing together the efforts of nuclear many-body theorists, quantum information theorists, and mathematicians in order to present and discuss algorithms for studying nuclear systems using recent progress in quantum information theory.

B. Structure of the school

We held approximately forty hours of lectures over three weeks and a similar amount of practical exercise and project work sessions, including the development of individual problems and the organization of exercises and projects. The website with all educational material is at [. The teachers of the school are planning to write, based on the material which was developed for the course, a Lecture Notes in Physics \(Springer Verlag\) monograph on *Quantum Computing for Many-body Physics*.](#)

The mornings consisted of lectures and the afternoons were devoted to exercises and project work meant to shed light on the exposed theory, projects and invited speakers.

The total load was estimated to correspond to approximately 170 hours, corresponding to **7 credits ECTS** in Europe. The days were organized as follows: 9.15-12.15 lectures, time for exercises with assistance (including lunch) till approximately 18 (3-4 hours of allocated exercise and project work sessions per day). The course was held at the premises of the ECT* (Villazzano) in Trento, Italy, from June 16 to July 4 in 2025.

The students were divided into groups from the very first day of the school. These groups remained stable throughout the whole duration of the course. Students collaborated and worked on exercises and projects through their respective groups. The groups were carefully put together to maximize diversity of backgrounds.

We organized several events (weekly social dinners as well) where individuals and groups exchange their experiences, difficulties and successes to foster interaction. The first two weeks we invited as speakers Justin Lietz from Nvidia (former MSU graduate student and participant at several Nuclear Talent schools) and Federica Mantegazzini, an experimentalist working on quantum technologies from the FBK foundation in Trento.

Five students handed in their final reports and requested credits.

C. Course Schedule

Lectures were approximately 45 min each with a small break between each lecture. The final schedule (with links to the educational material) is listed here.

D. Teachers and organizers

The teaching teams consisted of quantum computing theorists and nuclear theorists, with expertise ranging from many-body methods to quantum field theories. Several of us had developed courses on quantum computing and/or taught similar courses, in addition to our ongoing research on quantum computing and nuclear many-body physics.

The organizers and teachers were

1. Alexei Bazavov at Department of Computational Mathematics, Science and Engineering and Department of Physics and Astronomy, Michigan State University, East Lansing, MI 48824, USA. He is a theoretical particle physicist specializing in study of strongly coupled theories, in particular, Quantum Chromodynamics. Items of particular interest to him include quantum computing and algorithms, quantum field theory, finite-temperature field theory, lattice gauge theory with applications to particle and nuclear physics, parallel algorithms, iterative solvers, molecular dynamics algorithms, inverse problems and Bayesian inference, ultra-cold atomic systems and quantum simulations and effective field theory.
2. Zohreh Davoudi, Department of Physics and Center for Quantum Information and Computer Science (QuICS), University of Maryland, College Park, MD 20742, USA. Davoudi is an expert in lattice QCD for nuclear physics. She further specializes in quantum simulation and computing for quantum field theories, including lattice gauge theories and effective field theories of nuclear physics.
3. Morten Hjorth-Jensen at Facility for Rare Isotope Beams and Department of Physics and Astronomy, Michigan State University, East Lansing, MI 48824, USA & Department of Physics, University of Oslo, N-0316 Oslo,

Week and date	Time	Topic
Week 1 – Mon Jun 16	08:00–09:00	Registration
	09:00–09:15	Welcome by ECT* director Bira van Kolck
	09:15–10:00	Basic elements of quantum computing and review of quantum mechanics (Lecture material)
	10:00–10:15	Break
	10:15–11:00	Qubits, density matrices, and entanglement (Lecture material)
	11:00–11:30	Coffee break
	11:30–12:15	Quantum gates and quantum circuits (Lecture material)
	12:15–14:30	Lunch
	14:30–18:00	Exercises on Qiskit, gates, measurements, teleportation, T-injection (advanced) (Exercises)
Week 1 – Tue Jun 17	09:15–10:00	Review and introduction to quantum simulations (Lecture material)
	10:00–10:15	Break
	10:15–11:00	Measuring observables, Hadamard test, non-unitary gates (Lecture material)
	11:00–11:30	Coffee break
	11:30–12:15	Hamiltonian dynamics methods (Lecture material)
	12:15–14:30	Lunch
	15:30–17:00	Seminar: Quantum Computing at NVIDIA (Justin Lietz)
	14:30–15:30, 17:00–18:00	Exercises: time evolution, observables, Hadamard test (Exercises)
Week 1 – Wed Jun 18	09:15–12:15	Hamiltonian formulation of lattice gauge theories I–III (Lecture material)
	12:15–14:00	Lunch
	14:00–17:00	Exercises: lattice Schwinger model (Exercises)
Week 1 – Thu Jun 19	09:15–12:15	Quantum computation of gauge theory time evolution I–III (Lecture material)
	12:15–14:00	Lunch
Week 1 – Fri Jun 20	14:00–17:00	Exercises (Exercises)
	09:15–12:15	Quantum computation of lattice gauge theories IV–V and perspectives (Lecture material)
	12:15–14:00	Lunch
Week 2 – Mon Jun 23	14:00–17:00	Exercises
	09:15–12:15	Variational Quantum Eigensolver (VQE) I–III (Lecture material)
	12:15–14:00	Lunch
Week 2 – Tue Jun 24	14:00–17:00	VQE Exercises
	09:15–12:15	Nuclear Hamiltonians, Jordan–Wigner, Trotter evolution, observables (Lecture material)
	14:00–15:00	Exercise: Implement Trotter evolution (Colab)
	15:00–16:00	Seminar: Quantum Technologies in Trento (Federica Mantegazzini)
Week 2 – Wed Jun 25	16:00–17:00	Exercises
	09:15–12:15	Quantum Fourier Transform and Quantum Phase Estimation I–II (Lecture material)
	12:15–14:00	Lunch
Week 2 – Thu Jun 26	14:00–17:00	Exercise on QPE (Colab)
	–	ECT* closed (Feste Vigiliane)
Week 2 – Fri Jun 27	09:15–12:15	Quantum channels and errors; error mitigation; state preparation for the Schwinger model (Lecture material)
	12:15–14:00	Lunch
	14:00–17:00	Exercise on ZNE (Colab)
Week 3 – Mon Jun 30	09:15–12:15	Rodeo algorithm, adiabatic state preparation, nuclear physics applications I–III (Lecture material)
	12:15–14:00	Lunch
	14:00–17:00	Project work
Week 3 – Tue Jul 1	09:15–12:15	Rodeo algorithm IV–VI (Lecture material)
	12:15–14:00	Lunch
	14:00–17:00	Project work
Week 3 – Wed Jul 2	09:15–12:15	Modern techniques (LCU, QSP, QDRIFT) I–III (Lecture material)
	12:15–14:00	Lunch
	14:00–17:00	Project work
Week 3 – Thu Jul 3	09:15–11:00	Quantum error correction I–II (Lecture material)
	11:00–11:30	Coffee break
	11:30–12:15	Discussion of projects and summary of course
	12:15–14:00	Lunch
Week 3 – Fri Jul 4	14:00–17:00	Project work
	09:15–10:00	Discussion and work on projects
	10:00–10:15	Break
	10:15–11:00	Discussion and work on projects
	11:00–11:30	Coffee break
	11:30–12:15	Discussion and work on projects
	12:15–14:00	Lunch
	14:00–17:00	Project work and end of TALENT course

Norway. Hjorth-Jensen has his background in studies of different many-body theories applied to problems in nuclear physics and condensed matter physics. He works also on quantum engineering and machine learning applied to many-body systems. He has over many years developed introductory and advanced learning material in many-body physics, quantum computing, computational physics and machine learning.

4. Dean Lee at Facility for Rare Isotope Beams and Department of Physics and Astronomy, Michigan State University, East Lansing, MI 48824, USA. His research is focused on connecting fundamental physics to forefront experiments. He studies many aspects of quantum few- and many-body systems. Together with collaborators, he has developed lattice Monte Carlo methods that probe strongly-interacting systems and study superfluidity, nuclear clustering, phase transitions, and other emergent phenomena from first principles. He is also engaged in novel applications of new technologies for scientific research. This includes new algorithms for quantum computing and the development of emulators and machine learning algorithms based on concepts such as eigenvector continuation.
5. Ryan LaRose at Department of Computational Mathematics, Science and Engineering, Michigan State University, East Lansing, MI 48824, USA. He does research in computational physics and quantum information science. He is interested in both the physics of computation and the computation of physics - that is, what quantum physics can tell us about information and computer science, and how quantum computers can solve practical problems in physics and related fields.
6. Alessandro Roggero, Department of Physics, University of Trento, Povo, 38123 Trento, Italy. He works on simulations of strongly correlated quantum many-body systems using a combination of classical techniques and calculations on quantum devices. In particular, in understanding the role of entanglement in both observable properties of physical systems and as a fundamental tool to understand the structure of many-body states. His research interests include quantum simulations of inelastic nuclear processes, neutrinos in dense matter, quantum computing and algorithms, quantum Monte Carlo methods and effective field theory.

Morten Hjorth-Jensen and Alessandro Roggero functioned as student advisors and coordinators.

E. Budget

The ECT* sponsored all lunches and two social dinners for all participants, teachers included. Furthermore, all students and teachers got their housing expenses covered by the ECT*, except for students and teachers from the University of Trento. The FRIB-Theory alliance sponsored travel for several US participants. For the other participants, travel was covered by their respective home institutions.

F. Participants with their home institutions and nationalities

The target group for the Nuclear Talent courses was Master of Science students, PhD students and early post-doctoral fellows. Out of 52 applicants, 32 students were selected. Priority was given to Master of Science students and early PhD students.

The students were expected to have operating programming skills in C++ and/or Python and knowledge of quantum mechanics at an intermediate level, with some basic knowledge of many-body physics and also quantum computing. Sixteen different nationalities (participants and teachers included) and 24 different institutions/affiliations were represented. The students, their respective institutions and nationalities are listed in the table below.

On average, the students performed very well. It was an excellent and very active group. Five students handed in the final report and received a credit transfer of the seven ECTS from the University of Trento.

Name	Institution	Nationality
Cobos Jiminez, Jesus	University of Basque Country	Spain
Gallego Lizarribar, Kerman	University of Barcelona	Spain
Hsieh, Chung-Chun	University of Maryland	Taiwan
Heimsoth, Daniel	University of Wisconsin	USA
Jiang, Chongji	Beijing University	China
Andagana Rios, Pablo	Pontificia Universidad Católica de Chile	Ecuador
Gibbs, Joe	University of Surrey	UK
Brolli, Stefano	University of Milano	Italy
Nunziata, Gennaro	Universita della Campania Luigi Vanvitelli	Italy
Le Noan, Oscar	IPHC - CNRS - University of Strasbourg	France
Bofos, Stavros	CEA	Greece
Sarma, Chandan	University of Surrey	India
Salim, Nabeel	Indian Institute of Technology Roorkee	India
Dassie, Alan Cruz	GANIL	France
Merritt, Hosea	Ohio State University	USA
Azam, Clementine	CEA	France
Di Donna, Andrea	University of Trento	Italy
Restrepo-Giraldo, Alejandro	University of York	Colombia
Cariello, Nicholas	Michigan State University	USA
Mahendra Shinde, Rohit	Indian Institute of Technology Roorkee	India
Parini, Luca	University of Salento	Italy
Das, Ankit Kumar	Indian Institute of Technology Roorkee	India
Ghosh, Tanmoy	University of Zagreb	India
Sainato, Steven	CEA	France
Nigro, Luca	University of Milano	Italy
Covalam Vijayakumar, Nithish Kumar	TU Darmstadt	India
Sola Cava, Josep	Ruhr University Bochum	Spain
Urvashi, Urvashi	Indian Institute of Technology Roorkee	India
Jai, Prakash	Indian Institute of Technology Goa	India
Krueger, Raphael	Ruhr University	Germany
Kaur, Karmjeet	Akai University	India
Rani, Saba	University of Trento	Pakistan

II. BRIEF SUMMARY



The support from the ECT*, and its year-long experiences with running doctoral training programs and Talent courses, was central to the success of the course. And without the administrative help from Barbara Gazzoli and the staff at the ECT*, this school would not have been possible. Barbara was essential for the success of the school. Overall, we as teaching feel the original learning outcomes and goals were met. It was a marvelous group of students to teach.